

Salman Haider

GIS Analyst | ArcGIS & Python Automation

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Summary

GIS professional with a Master of Geographic Information Systems from the University of Calgary and practical experience in ArcGIS Pro, spatial analysis, map production, remote sensing, spatial data QA/QC, Python/ArcPy workflow automation, and AI-assisted geoprocessing. I have delivered 105+ client-facing mapping projects and worked with cadastral, demographic, satellite imagery, transit, and routing datasets. My strength is turning spatial data into clean maps, reliable workflows, and clear outputs that help teams make better decisions.

Skills

ArcGIS Pro • ArcGIS Online • ArcGIS Enterprise • ArcGIS Portal • ArcGIS Server • ArcGIS Field Maps • ArcGIS Dashboards • ArcGIS StoryMaps • ArcGIS Experience Builder • QGIS • Global Mapper • PostGIS • PostgreSQL • MS SQL • GeoServer • Leaflet • Mapbox GL JS • Mapbox Studio • MapTiler • Tippecanoe • Docker • Mango Maps • Python • ArcPy • SQL • JavaScript • HTML/CSS • Flask • GeoPandas • Shapely • GDAL • FME • Google Earth Engine • ENVI • ERDAS Imagine • Image Classification • Remote Sensing • Land Cover Classification • Spatial Statistics • Data Visualization • Raster/Vector Analysis • Network Analysis • OD Cost Matrix • Suitability Modeling • MCDA • Spatial QA/QC • Geocoding • Demographic Analysis • Territory Design • Cartography • Microsoft Office

Experience

Franchise Ready

Remote, Australia

GIS Coordinator

11/2021 - 08/2024

- Delivered **105+ franchise territory mapping** projects for clients across different industries, turning business briefings into GIS datasets, territory maps, reports, and web map deliverables.
- Worked from client briefs provided by the Director of Operations, reviewing each client's business model, competitors, target demographics, existing locations, sales patterns, and territory goals before starting the GIS work.
- Acted as the GIS point of contact during client follow-ups, explaining available datasets, mapping options, territory criteria, data limitations, and practical ways to fill gaps where suitable data was not readily available.
- Used ArcGIS Pro, ArcGIS Business Analyst, Australian Bureau of Statistics Census data, and demographic enrichment tools to build territory layers based on population, income, target demographics, competitor coverage, travel distance, and local market conditions.
- Created balanced franchise territories in ArcGIS Pro by combining client requirements with demographic patterns, competitor presence, existing sales information, POI access, and service-area coverage.
- Prepared client-ready cartographic products, territory summaries, project reports, ArcGIS Dashboards, and MangoMap web maps so clients could review territory boundaries and understand the analysis behind each recommendation.
- Automated repeatable project tasks with Python and ArcPy, including geocoding support, competitor and POI data preparation, spatial processing, map production steps, and quality checks to save time across recurring territory projects.

GreenAge Services

Islamabad, Pakistan

GIS Associate

04/2021 - 11/2021

- Created cadastral and land record maps using ArcGIS for government land record digitization projects.
- Worked under a GIS Specialist on a government land record digitization project, converting old scanned paper maps of state land into structured digital GIS datasets.
- Created and maintained geodatabases for assigned project areas, digitizing state land parcels, boundaries, and related cadastral features with close attention to geometry and attribute accuracy.
- Collected ground control points with RTK GPS and supported georeferencing of scanned maps, satellite imagery, and drone imagery for accurate project basemaps.
- Wrote Python and OpenCV scripts to read scanned PDF/JPG map records, identify region information, and organize files into the correct folders for GIS associates working on assigned areas.
- Developed Python QA/QC scripts to check geodatabases and shapefiles for incorrect, missing, or misspelled attribute values before final mapping use.
- Prepared GIS databases, map layouts, basemaps, and supporting spatial datasets for planning workflows, internal review, and project communication.

Education

 **University of Calgary** Calgary, Canada
Master's degree, Geographic Information Systems 08/2024 - 05/2026

- Graduate training in spatial analysis, remote sensing, cartography, spatial databases, urban GIS, web GIS, GIS project design, and Python-based GIS automation.
- Worked with ArcGIS Pro, ArcGIS Online, ArcGIS Enterprise, enterprise geodatabases, data versioning, Experience Builder, StoryMaps, advanced geoprocessing, and projection workflows.

 **Arid Agriculture University** Rawalpindi, Pakistan
Bachelor of Science - Geoinformatics and Earth Observation 09/2016 - 10/2020

- Undergraduate training in GIS, remote sensing, cartography, photogrammetry, geodatabases, spatial analysis, surveying, web GIS, and image processing.
- Final year project used GIS and remote sensing methods to identify suitable rainwater harvesting sites in the Potohar region.

Academic Projects

Geo Copilot: Natural Language GIS Tool for ArcGIS Pro

github.com/salman12haider13/arcgis-geo-copilot-llm

ArcGIS Pro | ArcPy | Python | LLMs | MGIS Capstone

- Developed a custom ArcGIS Pro script tool (Geo Copilot) that enables users to perform geoprocessing tasks using natural language queries, integrating **Large Language Models (GPT-5 and Llama 3.1)** with ArcPy workflows.
- Designed structured reasoning frameworks to translate user intent into executable spatial workflows.
- Evaluated model performance across 36 geospatial test scenarios, achieving up to **91.7%** function selection accuracy and **83.3%** parameter accuracy with GPT-based workflows, and analyzed failure cases related to projection conflicts and multi-step reasoning.

Transit-to-Table: Calgary Fresh Food Access

ArcGIS Pro | Network Analyst | OD Cost Matrix | Urban GIS | MGIS Project

- Built a **GTFS-based multimodal transit network** using Calgary Transit schedule data, street centerlines, ArcGIS Public Transit Data Model, and walking + transit travel-time costs.
- Modeled DA-level access from Dissemination Area origin points to fresh food destinations assembled from OSM grocery/supermarket points and Calgary food hubs using a 45-minute weekday OD cost matrix.
- Found **12 dissemination areas** with no access and **32 dissemination areas** with poor access. These areas represented **33,212 residents**, showing that Calgary has transit-based fresh-food access but that access quality is uneven.

Namal Lake Desiccation Assessment, 2000 to 2024

Remote Sensing | Google Earth Engine | Landsat 8 | ArcGIS Pro | MGIS Project

- Processed multi-temporal Landsat 7, Landsat 8, and Sentinel-2 imagery in Google Earth Engine to assess Namal Lake surface area change from 2000 to 2024, using MNDWI for annual water extraction and NDVI to estimate agricultural expansion around the lake.
- Integrated ERA5-Land climate variables including temperature, precipitation, watershed evaporation, and lake surface evaporation to build an annual environmental time-series dataset.
- Applied standardized OLS regression in Python (statsmodels) to evaluate the relative influence of agricultural and climatic factors on lake surface area change.
- Found a statistically significant decline in lake surface area over the study period, with agricultural expansion identified as the strongest negative predictor ($\beta = -0.711$, $p < 0.001$).
- Achieved an adjusted R^2 of 0.594 (model $p < 0.001$), indicating that localized agricultural expansion was a stronger driver of lake desiccation than the tested annual meteorological variables.

Brazeau Dam Land Cover Classification and NDVI Time-Series Analysis

Remote Sensing | Google Earth Engine | JavaScript | ArcGIS Pro | MGIS Project

- Built a 4,567 sq. km remote sensing study area around Brazeau Dam in Google Earth Engine and processed Landsat 5, 7, and 8 imagery for land cover classification and NDVI-based vegetation change analysis.
- Created both unsupervised and supervised land cover classifications using K-Means clustering, CART, training/validation samples, class remapping, and confusion matrix accuracy assessment.
- Achieved 71.875% overall classification accuracy and built 1990, 2005, and 2020 NDVI median composites, showing vegetation decline by 2005 followed by recovery trends by 2020.

Academic Projects

GEOSS Remote Sensing MVP Workflow for Vegetation Risk and Property Maintenance

Remote Sensing | Postgresql | PostGIS | Mapbox | Segment Anything Model | Landsat 8 | ArcGIS Pro | MGIS Project

- Designed and presented a remote sensing MVP workflow for aerial vegetation identification, property risk assessment, and maintenance planning.
- Used a sampled Vancouver study area to demonstrate the workflow logic rather than claiming a full production deployment.
- Proposed a pipeline using high-resolution orthophotos, Microsoft building footprints, PostGIS storage, Mapbox visualization, AWS processing, Python, QGIS, and SAM-based segmentation.
- Designed a building segmentation workflow using building footprints, 20-foot buffers, clipped orthophoto tiles, samgeo box prompts, and GeoJSON outputs stored in PostGIS.
- Designed a tree extraction workflow using gridded orthophoto tiles, samgeo segmentation, and GeoJSON tree polygon outputs.
- Designed a canopy proximity workflow to calculate building area covered by tree canopy and count trees within a defined buffer distance.
- Designed a moss detection workflow using multispectral imagery, building polygons, tree canopy removal, clipped building areas, and NDVI thresholding.
- Framed the workflow around practical risk use cases such as tree failure risk, moss growth, property damage, insurance risk, and municipal planning.

LiDAR and Object-Based Conifer Tree Height Classification

Remote Sensing | ENVI 6.0 | LiDAR Canopy Height Model | ArcGIS Pro | MGIS Project

- Used ENVI 6.0 and ArcGIS Pro to classify land cover and coniferous tree heights in a Campbell River, British Columbia study area.
- Processed a 1 metre LiDAR Canopy Height Model to classify coniferous trees by height.
- Used QuickBird imagery with NIR, RGB, and NDVI bands for example-based feature extraction.
- Tuned image segmentation parameters using scale level and merge level to improve object separation.
- Created training samples for roads, big trees, small trees, and clear-cut areas.
- Tested KNN and SVM classification methods and selected SVM because it performed better for identifying small clear-cut patches.
- Used rule-based classification on the LiDAR CHM to create nine tree height classes from 0 to 45 metres.
- Found the largest conifer height class was 25 to 30 metres, covering 307,565 square metres or 13.67 percent of the study area.
- Found the 20 to 25 metre height class covered 278,472 square metres or 12.38 percent of the study area.
- Found very tall conifers were rare, with the 40 to 45 metre class covering only 250 square metres or 0.01 percent of the study area.

Logistics Routing MVP

github.com/salman12haider13/Logistics-Routing-MVP-Demo

PostGIS | pgRouting | Flask | Docker | Leaflet | OpenStreetMap | MGIS Project

- Built a self-hosted logistics routing MVP to test how route planning could work without relying on external routing APIs, using OpenStreetMap road data, PostgreSQL/PostGIS, pgRouting, Flask, Docker, and Leaflet.
- Designed the full routing workflow from raw OSM .pbf data to a routable PostGIS network, including road filtering, topology creation, source/target nodes, distance and travel-time cost fields, one-way road handling, and spatial indexes.
- Developed a Flask backend API that accepts start/end points, routing mode, blocked polygons, traffic points, and number of route alternatives, then returns route geometry and summaries as GeoJSON for the web map.
- Created an interactive Leaflet routing console where users can place start and end points, draw blocked zones, add traffic-delay points, compare route alternatives, and review distance, travel time, and route steps.
- Added blocked-zone and traffic-point logic to make the prototype closer to real logistics planning, where routes may need to avoid restricted areas, account for delays, and compare shortest-distance versus shortest-time options.